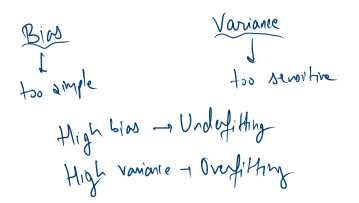


training accuracy = 99%
 test " = 75%



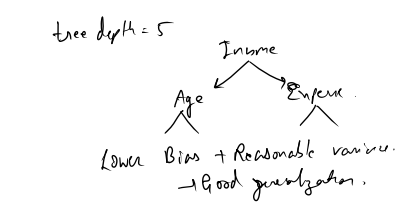
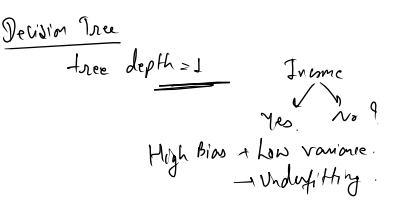
Bias	Variance	What happens?
High	Low	Underfitting
Low	High	Overfitting
High	High	Bad
Low	Low	Ideal model

Simple model → more complex → more complex → very complex

Bias decrease

Variance increase

Bias-variance trade off.



Regularization:

Model	Model	Normalizing
$w_1 = 2$	$w_1 = 100$	Loss = prediction error.
$w_2 = 3$	$w_2 = -200$	Regularized loss
$w_3 = 4$	$w_3 = 150$	Loss = pred. error + penalty

Ridge / L_2

Large coefficients
 ↓
 Shrink them

Lasso / L_1

Large coefficients.
 ↓
 Shrink them

Ridge

MSE + $\lambda \sum_{j=1}^p \beta_j^2$ → regularization strength

$\lambda = 0$

↑ λ

$\text{max} = \text{MSE}$

Lasso

MSE + $\lambda \sum_{j=1}^p |\beta_j|$

$\beta_j = 0$

Elastic Net = $L_1 + L_2$

MSE + $\lambda_1 \sum |\beta_j| + \lambda_2 \sum \beta_j^2$

